Don't worry, this is not a real test! This is just a collection of problems that lets me know what you know. If you dont know how to do a question, feel free to leave it blank or write down initial thoughts

- You can use any reference you want
- only use a calculator on questions with (🖩) on them
- keep track of the time you spend on each question

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1. Set and interval notation
   (a) Translate into interval notation: ”real numbers greater than -2 & less than or equal to 5”

   (b) Translate into set notation: ”integers greater than -2 & less than or equal to 5”

   (c) Translate into interval notation: ”real numbers less than or equal to -2 or bigger than 5”

   (d) Translate into set notation: $(-\infty, 3) \cup (3, \infty)$

   (e) Try translating (c) into set notation (A bit trickier)

2. Basic pseudocode: What value does x have after executing the following pseudocode?

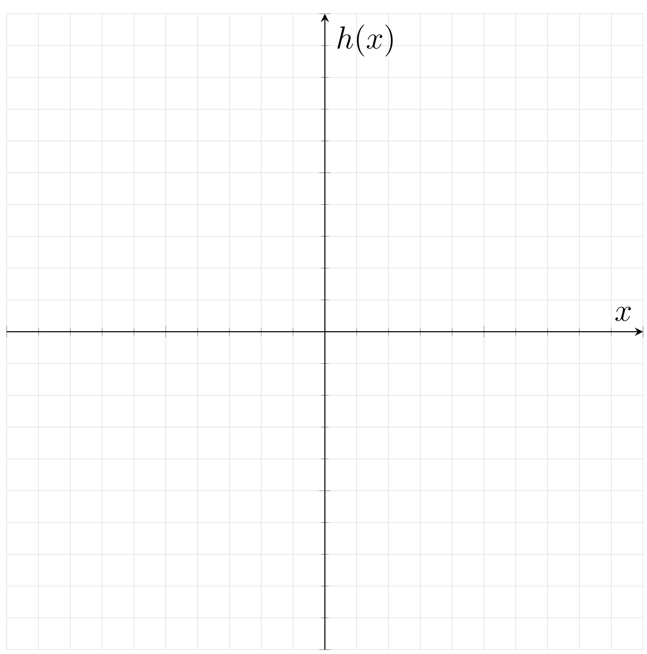
```
x <- 0 # this means let x take the value 0
while x < 10 do
    if x is even then
        x <- x + 3 # remember that 0 is even :)
    else
        x <- x + 1
```

3. Let $f : \mathbb{R} \to \mathbb{R}, f(x) = x^3$ and $g : \mathbb{R} \to \mathbb{R}, g(x) = 2(x-3)^3 - 3$

   (a) Find a series of transformations that takes f to g

   (b) Are there any points of intersection between f and g? If so, find them. ( )

   (c) Plot the graph of $h(x) = \frac{f(x)}{x} + 2x$ and label key points



(d) Find the slope of a tangent line to g at $x = 3$ and $x = 4$

(e) Find the *signed* area under the curve g from $x = 3$ to $x = 6$

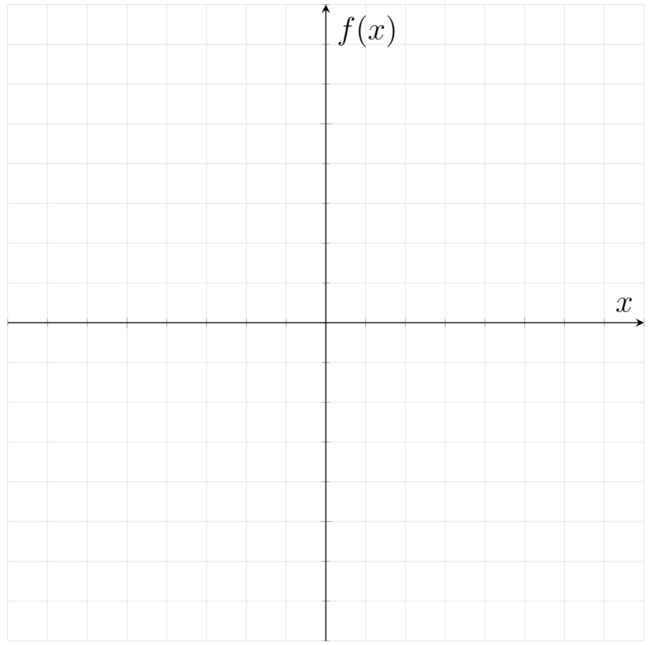
(f) Find the *unsigned* area between the curve g and the x axis from $x = 3$ to $x = 6$

4. Let $f : [0, \frac{13\pi}{12}) \to \mathbb{R}, f(x) = \sin(2x) + 3$

   (a) Find the period and amplitude of f

   (b) Find the endpoints of the graph of f

(c) Sketch the graph of f over its domain, labelling key points



5. You have a sheet of paper with thickness of $0.1mm$. You keep folding it in half, and measure the thickness after each fold.

   (a) What is the thickness after 3 folds?

   (b) let $f(n)$ be the thickness in mm after n folds. Write the function f in terms of n

   (c) The moon is $384400km = 384400 \times 10^6 mm$ away from the surface of earth. How many folds do you need for the paper to reach the moon?

6. Every morning before school, you pick a pair of socks at random from your drawer without looking. Suppose you have 3 pink, 8 white, and 7 black socks in the drawer.

   (a) What is the probability of picking a pair of matching white socks? (Hint: try a tree diagram)

   (b) What is the probability of picking a pair of matching socks (any colour)?

   (c) Given you chose a pair of matching socks, what is the probability that you chose a pair of black socks?

7. Counting: Now dont worry if you dont know how to do these, there is a lot of variability in how different schools cover this topic (if they cover it at all) but having a decent understanding of these concepts can be extremely useful in probability/stats topics later on.

   (a) How many different ways can you arrange the letters in the word "METHODS"? (🖩)

   (b) Explain what $\binom{10}{4}$ means and how do you calculate it? Explain why the formula for $\binom{n}{k}$ makes sense